

CLASH OF THE **CLAWS**

MSRIT | TEAM NEXUS

**NEXUS: Local sovereign
agentic AI Assistant**

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THE PROBLEM: Privacy vs. Productivity



The Overwhelmed Professional

Juggling tight schedules, fragmented apps, and constant context switching across 5+ platforms daily.

Cloud-Dependent AI

Current assistants like Siri and Google Assistant rely on cloud servers, creating privacy vulnerabilities with every query.

No Cross-App Orchestration

Can't connect the dots between a late ETA and your Zomato order, or between a calendar change and Slack notification.

The Context Switch Tax

Manually updating each app when plans change. Why should your data leave your phone just to send a text?

The Agentic Gap

1

Existing Tools

Siri, Google Assistant, and Cloud LLMs

- Reactive, not proactive
- Privacy risks
- Limited action capability

2

The Gap

No local orchestration layer

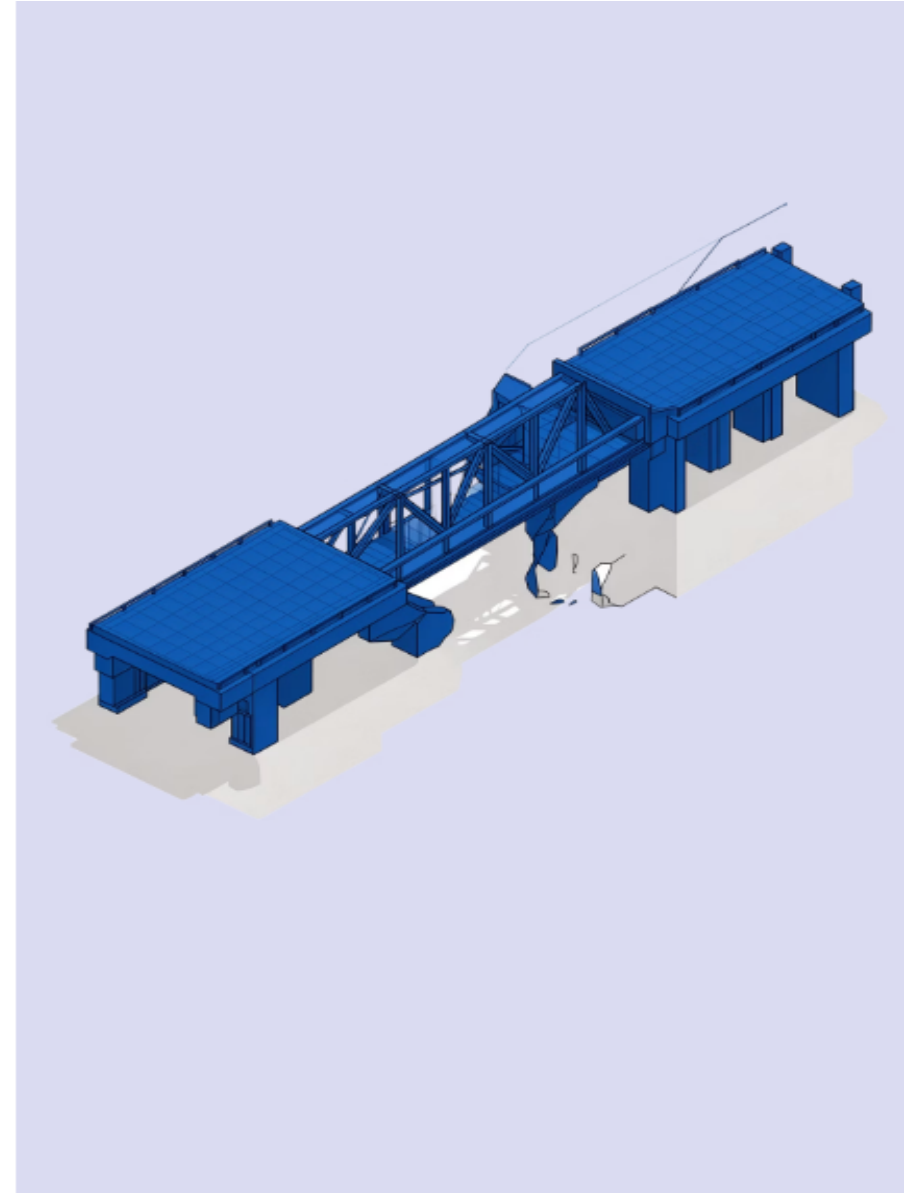
- Can't chain actions
- No cross-app intelligence
- Cloud-dependent reasoning

3

The Opportunity

On-device "Brain" that owns execution flow

- Local-first sovereignty
- Task contracts
- Autonomous execution



NEXUS: The Solution

A sovereign agent that thinks (Brain) and acts (Hands) via a strict Task Contract, executing across your local ecosystem without cloud prying.



Local Intent Parsing

Zero-data leakage. All reasoning happens on-device using local LLMs.



Cross-App Chaining

One trigger initiates multiple coordinated actions across WhatsApp, Calendar, Zomato, and more.



Human-in-the-Loop

Safety checks for high-cost or sensitive actions ensure predictable, safe execution.



Strict Contract Validation

JSON schema ensures every action is predictable, validated, and auditable before execution.

Product Demo: The "Late" Scenario

01

Trigger Detected

User says "I am late"

02

WhatsApp DM and Calendar

Automatically messages meeting lead with delay notification

03

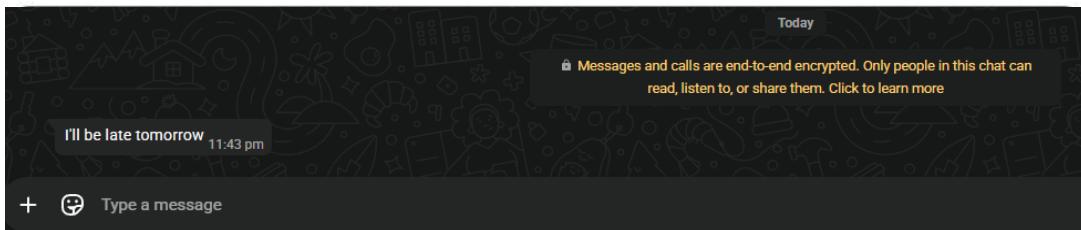
Calendar Update

Reschedules Weekly Sync by 20 minutes

04

Mapping and Privacy

Nexus protects your privacy by using local mapping to send only anonymized intents to Groq, ensuring your complete personal data never leaves your device.



Technical Architecture



OS Layer

Local storage, permissions, and app integrations



NEXUS Brain

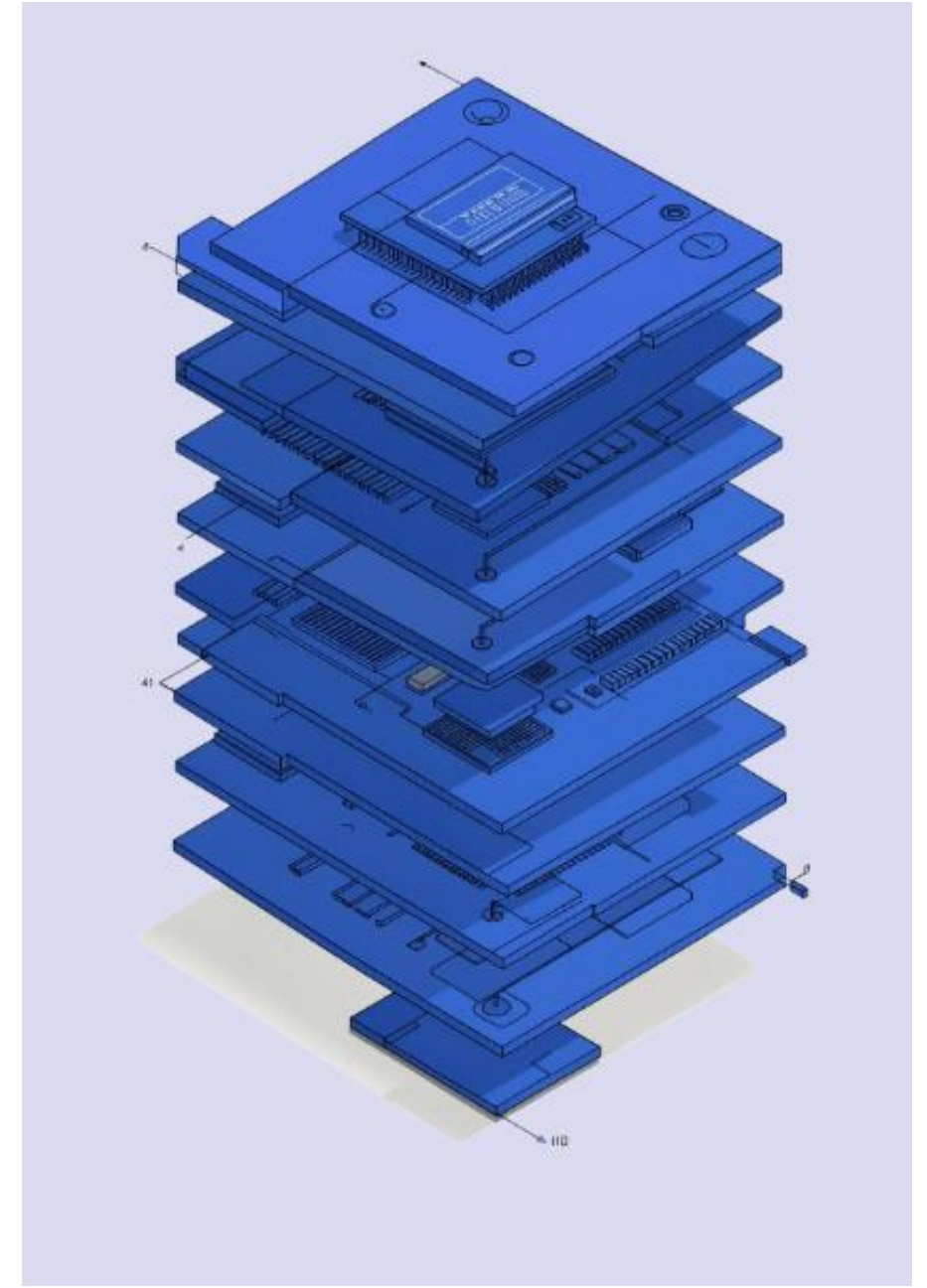
Node.js runtime with local LLM for intent parsing and Task Contract generation



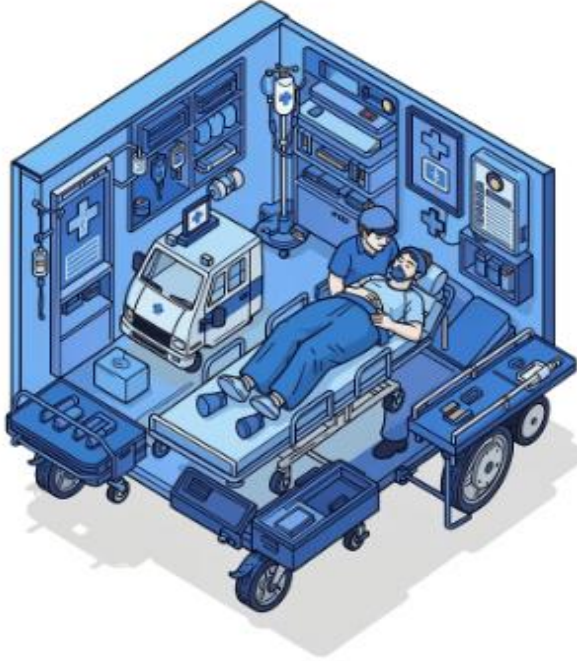
App Connectors

API integrations with WhatsApp, Zomato, Calendar, and other services via validated Task Contracts

Tech Stack: Node.js, AJV for JSON Schema validation, local LLM for on-device reasoning, and REST APIs for app integration.



Impact & Use cases



Emergency Handling

"I had an accident" → Calls ICE, sends GPS location, cancels dinner reservation automatically.



Daily Routine

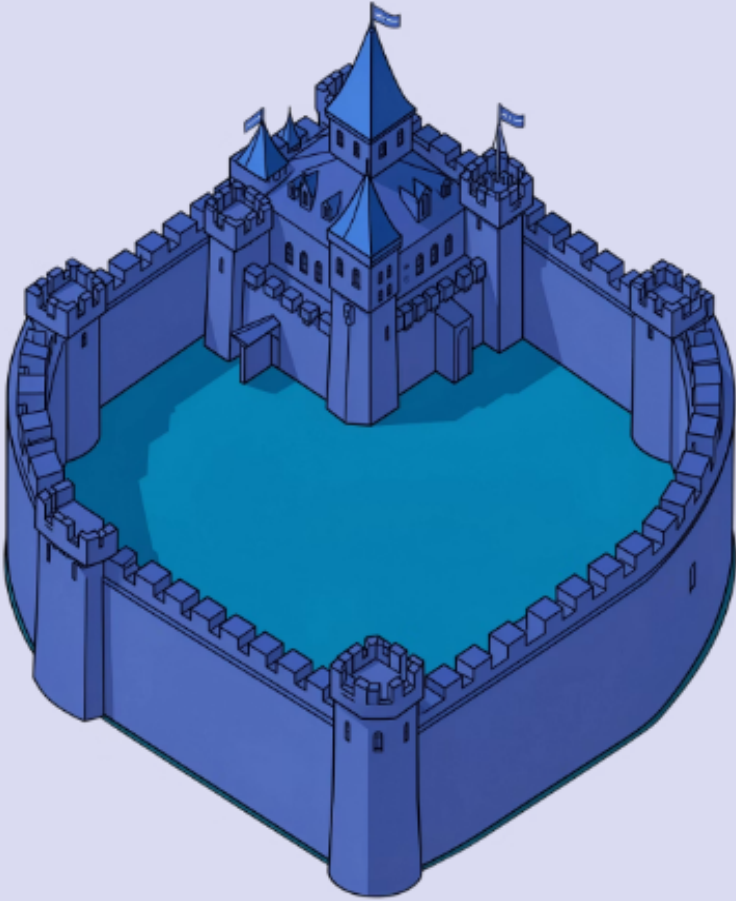
"Starting my workout" → Puts phone on Do Not Disturb, starts Spotify playlist, orders protein shake via Zomato.



Home Automation

"Leaving home" → Locks doors, turns off lights, adjusts thermostat, starts security cameras.

Extensible: Can be extended to any app with an API or intent-filter. The modular architecture makes integration simple.



The MOAT: Competitive Advantage

Local-First Sovereignty

NEXUS doesn't just suggest—it **executes** across your local ecosystem without cloud prying. Your data never leaves your device.

Strict Contract Validation

Unlike open-ended agents, NEXUS uses JSON schema to ensure every action is predictable, safe, and auditable before execution.

Human-in-the-Loop

Critical actions requiring approval (e.g., payments, sensitive messages) trigger user confirmation, maintaining control.

Documentation & Repository

1

GitHub Repository

<https://github.com/sinchana0901/Nexus>

2

Demo Video

https://drive.google.com/file/d/1QmbITxJNndy5jC6uYAlDgjUfOdq59dsE/view?usp=drive_link

3

Technical Documentation

https://drive.google.com/file/d/1-Slzq70d0fhWXy_KoadNzgVeL_ytfdnO/view?usp=sharing

Soul Constraints: Ethical Execution



Adjustable Constraints

Our agent respects user-defined boundaries through configurable constraints:

- `cost_under_2000`: Maximum spending limit
- `no_mass_messaging`: Prevents spam
- `approval_required`: Critical actions
- `privacy_level`: Data sharing limits

This ensures the AI doesn't just act, but acts **within the user's ethical and financial boundaries**.